

SSHVisual: Visual Tool to Analyze SSH Data

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Abstract

Network operators use visualizations to detect attacks on their systems, exploring and interacting with data to uncover patterns that would otherwise stay hidden. Operators then use these findings to block attackers and to expand the knowledge base of known threats. Detecting anomalies and uncovering deeper patterns iteratively in data is at best, tedious and at worst, not achievable without data visualizations. Previous works in network visualization analyzed the IP and AS dimensions of SSH logins without much focus on the username aspect of the attackers' attempts, detected anomalous network events only where baseline events were available, and enabled operators to share findings with one another. Little attention has been paid to whether attackers observed at different times represent the same persistent actor. We present an iterative visualization SSHVisual, using login attempts collected on our own SSH-facing systems. SSHVisual analyzes the IP, AS, and username dimensions of SSH attacks, in service of SSH campaign detection. It complements automated detection by allowing operators to find anomalous events without normal baseline events. It correlates SSH attacks with real-world events, and it introduces a similarity metric that quantifies attacker persistence over time. Like previous systems, SSHVisual supports iterative and collaborative sharing of findings between operators. We evaluated SSHVisual using case studies and found that it helps us block attacks and uncover patterns. We identified campaigns in which a set of attempted usernames recurred across unrelated IPs and ASes. The similarity metric surfaced attackers returning across different time periods despite changing network origin. SSHVisual shows that treating usernames as a first-class analytic dimension reveals attacker coordination that IP-centric analysis misses, and that exploratory, operator-driven detection remains valuable.

I Introduction

Computer systems worldwide employ SSH [1] to allow their users to securely connect to them in order to execute commands and do remote administration. A lot of users do a username, password combination (password based authentication) in order to gain entry to the SSH system. Attackers, who are adversarial in nature, can use brute force attack mechanisms [2] to get into SSH systems. Sometimes, they guess what the username, password combinations are and other times, they take advantage of poorly secured systems to build botnets [2] to carry out further attacks [3, 4] or steal resources from the

systems, like cryptomining [5, 6].

Given the prevalence of SSH attacks in the IPv4 space [2], we collect SSH login data on our own systems and present SSHVisual. SSHVisual serves two kinds of network operators. Network administrators, who handle day-to-day SSH traffic, need to identify attackers quickly and block them. Security analysts, who study traffic over longer time periods, look for patterns that are not immediately apparent and share what they find across organizations. Consider a single time period: an administrator can isolate the highest-volume attacker and block it immediately, while an analyst examining the same window can iteratively surface a set of low-frequency attackers that attempt an identical sequence of usernames, and group them as a coordinated botnet.

Surveys of network visualization [7, 8, 9, 10, 11] establish the maturity of the field and the gaps we target. Our contributions follow from treating the username as an attacker identity. First, actor-level analysis: we treat the attacker as a first-class actor — including the username identity, not only the source IP or autonomous system (AS) — and we identify coordinated campaigns. Since network data affords no normal baseline for automated detection [12], this identification is necessarily exploratory (therefore, not antagonistic but complementary to automated detection with normal baselines) — operators recognize coordination visually rather than specifying it in advance. Second, a similarity metric that quantifies attacker similarity and lets operators track how actors evolve in the username dimension across time periods. Third, correlation of actor behavior with real-world events, which is more tractable through usernames than IPs, given the limitations of IP corroboration. Additionally, following established practices, we allow for iterative operator driven collaboration and labeling of irregularities found in the data.

The rest of the paper is as follows: Section II is Related Work, Section III is Tasks and Requirements, Section IV is Design, Section V is Evaluation and Section VI is Conclusion and Future Work.

II Related Work

A long line of tools [13, 14, 15, 16, 17, 18, 19, 20] render network traffic and security events navigable for operators and analysts with the majority being network flow and traffic oriented. Additionally, a smaller alert- and log-oriented group [21, 22, 23, 24, 25, 26, 27, 28, 29, 30] organizes IDS output and heterogeneous logs of which [23] contributes a visual analysis (of the CICIDS2017 dataset) that isolates at-

tack clusters — including SSH brute-force (SSH-Patator) — through clustering and dimensionality reduction, namely a multi-resolution wavelet-feature analysis.

A Attack Campaigns with Time

Honeypot literature does not promote the username identity as an actor in its own right (for attack campaign detection purposes): it remains an attribute to count, never a unit to track — the unit that, alongside the IP and AS, anchors our analysis. Some papers [31, 32, 33, 34, 35, 36, 37] collect and chart honeypot traffic, with several of them capturing SSH credentials (usernames and passwords). Where credentials are charted — the SCADA SSH honeypot [34], kippo-graph [35], and the container deployment [36] — usernames and passwords appear only as static top-N tallies (kippo-graph’s top-10 username and password bars are the archetype), never placed on a time axis even where the same tool charts attack volume over time. The WAN honeypot framework [38] is the closest prior system on the credential axis: it computes an IP-to-IP dictionary-overlap (containment) score to probe whether six same-/24 source IPs share a wordlist — a manual, one-off prototype of credential-based coordination analysis, performed on the IP unit and without a temporal dimension.

B Attacker Similarity with Time

Computed similarity between entities is well established in literature [39, 40, 41, 42]. We do not claim similarity computation itself as novel; the contribution lies in the entity it operates on and how it moves through time. On the entity: these systems cluster internal hosts [39], behavioral profiles [41], IP/port members [42], or internal enterprise users grouped by shared resources — not external attacker identities [40]. None treats the username as an attacker actor. On time: [41, 42] reach into the temporal dimension, but both do not compute the similarity of the attacker population across consecutive windows. We apply similarity based metrics — for example, Jaccard overlap of attackers between successive windows — particularly to the username actor unit, capturing how the attacker population persists or churns over time.

C Limits of automated detection

One premise of our approach is that SSH honeypot traffic does not admit the normal baseline that automated anomaly and campaign detection presuppose. The broader literature already documents the fragility of such detection: Sommer and Paxson [43] note that machine-learning anomaly detection is rarely deployed for intrusion detection given the variability of “normal” the cost of errors, and the difficulty of sound evaluation, while Axelsson [44] shows it is throttled by the base-rate fallacy, where the false-positive rate becomes the limiting factor once intrusions are rare relative to benign traffic. Our setting departs from that one entirely, in the direction Gates and Taylor [12] anticipate when they question whether attacks are rare or a stable normal exists, note that normal traffic may itself

be malicious and malicious traffic unremarkable, and observe that detection ultimately devolves to an operator investigating clusters of activity by hand. The consequence is visible in the systems cited above: Situ [39] learns a probabilistic model of each IP’s behavior and CLIQUE [45] scores each actor against a baseline of its own expected activity, so both presume a per-actor “normal” that near-uniformly hostile, constantly churning traffic cannot supply. Situ nonetheless concedes that automation alone is insufficient — operators “do not want another automated tool with algorithms they do not trust” — and builds a human-in-the-loop system; we reach the same conclusion from a starker starting point, lacking even the per-actor baseline it assumes, and so treat campaign identification as operator-driven from the outset.

D Actor identity and real-world event correlation

The attacks we observe do not arrive at a steady rate; they rise, fall, and change in character as events unfold in the wider world, so understanding them means lining them up against what is happening outside the network. The link is well documented: Bilge and Dumitras [46] found that the number of attacks against a given vulnerability can jump by up to five orders of magnitude — a hundred-thousand-fold increase — once it is publicly disclosed, and Sabottke et al. [47] showed that future exploitation can be predicted from signals in disclosure channels alone. Allodi and Massacci [48] add that a vulnerability’s official severity score is a poor guide to whether it is actually attacked, so the dependable way to learn what attackers are reacting to is to watch real activity and line it up against real events, not to read it off a score. We want to do this at the level of a campaign — a group of attackers acting together — rather than one attacker at a time, and that raises a practical problem: a campaign rarely shows up as a single source address. IP addresses do not stay tied to one machine — one host is assigned many addresses over time [49], hosts cycle through new addresses [50], and counting by address can overstate the number of real machines by orders of magnitude [51] — so it is likely that a view organized by address scatters one campaign across dozens of entries that look unrelated. The username pulls those pieces back together: the same login names likely recur across the changing addresses, so grouping by username re-assembles the campaign that an address-by-address view splits apart. A username is not a perfect identity either — names like “root” are tried by almost everyone, and one campaign runs through a whole list of names — so we treat it as one analysis dimension alongside the address and the network it belongs to (its AS), resolving the overlaps with the similarity and grouping methods above.

E Operator labeling and collaboration

Since detection on our data is operator-driven, the findings an operator produces by hand must be captured and shared rather than discarded. In two lines of work [52, 19], interactive labeling has been used to fold the operator’s judgments back into

the system. Collaborative analysis has been supported through shared artifacts with [17] and [20] being notable examples. These systems establish that labeling and collaboration are valuable, which is why our labeling-and-sharing mechanism is the step that turns an identified set of usernames into a durable, shared campaign.

III Tasks and Requirements

The design of SSHVisual derives from our(authors) experiences as network operators and as mentioned, the gaps and established practices that we have found in network literature. Accordingly, we formulate the tasks and requirements to accommodate network operator needs, fill out gaps and respect establish practices.

The visual tasks for network operators are as follows: Find SSH campaigns, using data that has Username, IP, AS, and other information, and share that information(T1); Find anomalies from data, block them and share that information(T2); Allow iterative collaboration based on visualized information(T3). T1 is aimed primarily at security analysts, T2 is aimed primarily at network administrators and T3 is aimed at both analysts and administrators.

The first requirement (R1) is profiling attacker similarity. The visualization should be able to algorithmically cluster similar attackers based primarily along the username dimension, secondarily along IP and AS information and tertiarily along other dimensions. R1 mostly serves T1 as it allows analysts to build a repository of similar attackers.

The second requirement (R2) is sorting of variables. The visualization should be able to show attackers who fit constraint(s) on variables as specified by the operator. R2 primarily fulfills T1 and T2 because administrators can sort out anomalies and analysts can find similar attackers given constraints.

The third requirement (R3) is visual clarity. Line charts show 10 attackers (for readability) at most (unless the operator wants to see more) using username, IP and AS along a temporal dimension. At the same time, an aggregate graph shows all the attacks without specifications. For instance, if 150M attacks happened on a given day in the aggregate graph and 75M of that came from one username/IP/AS in the username/IP/AS graph, those attackers can be identified easily. R3 primarily fulfills T2 as it allows for easier detection and eventual blockage of attackers.

The fourth requirement (R4) is tagging. Any aberrant attacker can be tagged by the network operator so that operators on the same network do not have to identify the same patterns/anomalies twice. Attackers tagged by one network operator are on display for other operators to see (as in, tagging attackers and making them stick to the visual). R4 supports T3 as it allows for collaboration.

The fifth requirement (R5) is exploratory workflows through different visualization types because no single visualization serves every need. The visual system should allow for sorting and clustering using tabular visualization, and linechart visual-

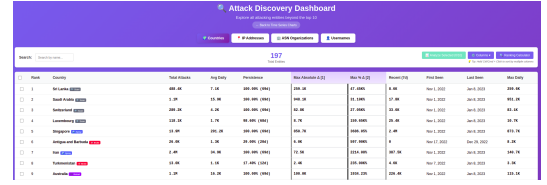


Figure 1: Tabular Viz - Requirement R5

ization based on clustering, and sorting. A tabular visualization is good for precise lookup of related attackers without a temporal structure and a linechart visualization lacks precision but preserves the temporal structure in order to show things like spikes and campaigns at a glance. For instance, an administrator can find an anomaly easily using filtering in the tabular visualization and may not have to see that in a linechart to block attackers. However, an analyst may need to use the tabular visualization to build a repository of similar attackers and use the linechart visualization to see those attackers visually. R5 supports T1 and T2 because two different people can use the same system for different means and to achieve different results.

IV Visual Approach – I realized too late that part of R5 is redundant and can be combined with R2 – Basically R5 should just be about why we need 2 visualizations and everything else from R5 can go into other requirements but mostly into R2 – Additionally, being able to drill down on the linecharts can be a requirement as well

For the user interface, we first choose to show the overall picture of what is happening in the data. So, accordingly, we show 2 views where the first view shows tabular visualizations that are temporal independent and the second view shows linecharts that are temporal dependent. By default, the tabular visualizations and the linecharts show data for all the time periods with the operator having the option to specify a narrower time period in the linecharts. The tabular visualization is shown in Figure 1 and the linecharts are shown in Figure 2. Additionally, in figure 1, the operator has the option to switch between tabs; currently, the tab is set to attackers along the username dimension but can be switched to countries, IP addresses or ASN organizations with all the other dimensions showing columns that can be similar or different. For instance, the country tab has a 'Unique ASNs' column which the ASN tab lacks.

Now, Figure 1 also illustrates the strength of a tabular visualization that a linechart visualization lacks. It helps operators

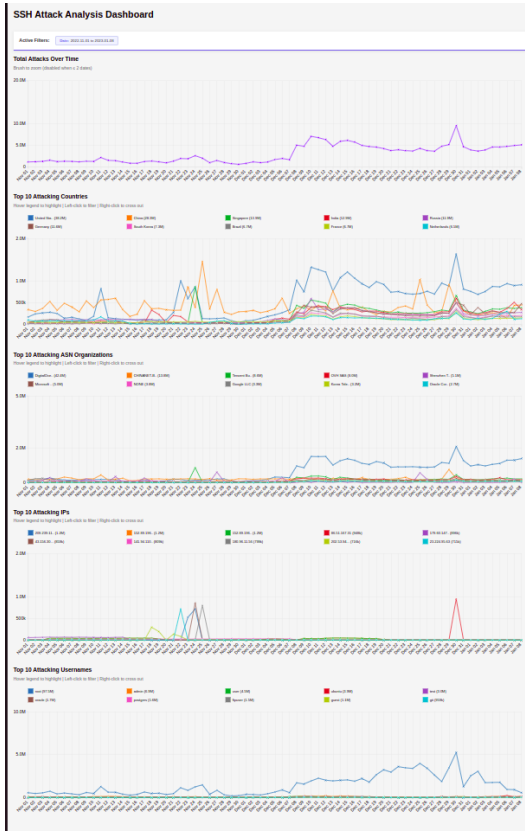


Figure 2: Linechart Viz - Requirement R5

sort and rank a dimension (in the figure: country) along column variables. This becomes especially important when the data is tail-heavy, that is, most of the data is dedicated to a few actors and almost everything else is buried beneath. However, and as mentioned, the tabular visualization does not help the operator answer questions along the temporal dimension. This is where Figure 2 becomes important. For instance, an operator can ask questions like, "Do the countries in Figure 1 have the same peaks and valleys in time?" or "Does one country attack earlier and another country attacks later?" and the tabular visualization will not be able to answer those question but a linechart visualization can do that. From Figure 2, the operator can see that there was an increase in activity on Dec 30 using the "Total Attack Over Time" chart and immediately also see that every country in the "Top 10 attacking countries" chart also had a spike in activity on Dec 30. Figure 1 and 2 show a visual system that fulfills requirement 5.

In figure 3, we see who the attacker base using the same usernames is. We do this by clustering how many distinct addresses, networks, and countries try a given username, how stable that population is day to day, and how concentrated the traffic is on its single most active source. Additionally, unlike IP addresses, where we run k-means clustering on seventeen dimensions/features, we do not do that for usernames because previous diagnostic analysis showed that some features were basically unusable. For instance, most of the IPs have at least one attack per day and therefore, they have a stability score

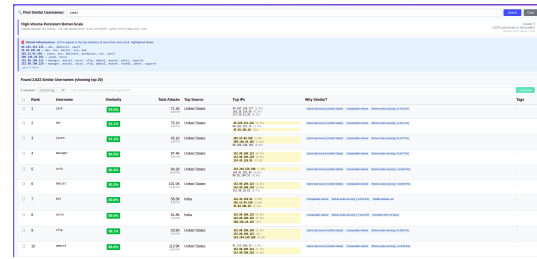


Figure 3: Username Similarity - Requirement R1

(which is a feature) that factors this feature into clustering. However, half of the usernames we saw do not have this feature (that is, they were only seen one day and therefore, calculating stability is meaningless) and therefore, it did not factor into the k-means clustering that we did for the username. Regardless, we saw nine clusters and we can see one of those clusters in Figure 3.

In Figure 4, we characterise IP addresses by a seventeen-dimensional behavioural profile drawn entirely from its observed activity over the time period in the data. The features fall into four groups: volume (log-scaled totals, daily means, and peaks), temporal shape (the fraction of days on which the address was active, its burst intensity, its largest day-over-day swings, a fitted trend slope, its activity span, and how much of its traffic falls in the final week), targeting behaviour (how many distinct usernames it tries, how stable that set is from one day to the next, its rotation rate, and the share taken by its single most-tried credential), and two flags for whether the address belongs to a cloud provider and whether it originates from a major source country. Then, k-means clustering is used to find similarity between the IPs. Each cluster is then automatically labelled from its centroid — "High-Volume Persistent Focused", say — so an operator sees an interpretable profile. While querying, finding addresses similar to a given IP means retrieving its cluster and ranking the other members by Euclidean distance.

Figure 3 and Figure 4 illustrate how our system helps operators compile a repository of similar attackers along multiple dimensions and satisfy requirement 1 accordingly.

[FOr Figure 5, I don't know what to write because I think that R5 covers that already]

[For figure 6, should I bring in references as to why having visual clarity is important?]

Figures 7,8, 9, and 10 show the process in tagging a similar group of attackers so that other operators using the visualization can see work that's already been done. Before figure 7, the operator observes a set of usernames acting very similarly. In Figure 7, they open the tagging section, in Figure 8, they create a new tag, in Figure 9, they tag a bunch of usernames that bear similarity to another username and in figure 10, the operators can see which of the usernames have already been tagged. These 4 figures support requirement 4 as they allow operators to collaborate and not do redundant work.

IP	Similarity	Metadata
192.168.1.1	0.95	192.168.1.1
192.168.1.2	0.92	192.168.1.2
192.168.1.3	0.88	192.168.1.3
192.168.1.4	0.85	192.168.1.4
192.168.1.5	0.82	192.168.1.5

Figure 4: IP Similarity - Requirement R1

Category	Value	Unit
Category 1	100	100%
Category 2	200	200%
Category 3	300	300%
Category 4	400	400%
Category 5	500	500%

Figure 5: Sorting - Requirement R2

V Implementation – how we made the tool. this section may not exist by the end

VI Evaluation and Discussion

A Case Studies

B Discussion

VII Summary

A Limitations

B Future Work

VIII Acknowledgments

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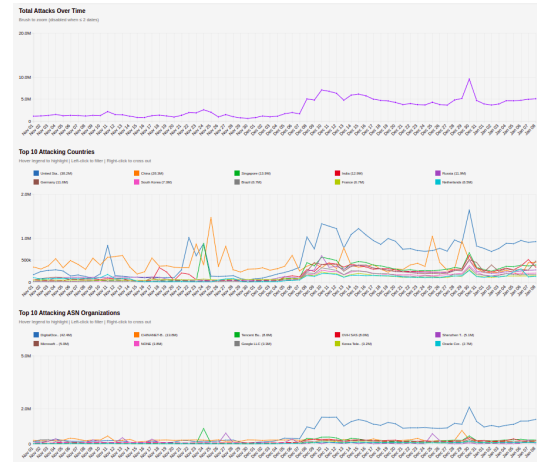


Figure 6: Visual Clarity - Requirement R3

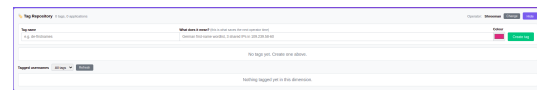


Figure 7: Tag Screen - Requirement R4



Figure 8: Creation of new Tag - Requirement R4

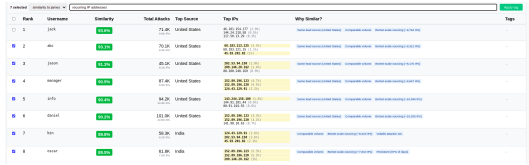


Figure 9: Addition of usernames to tag - Requirement R4

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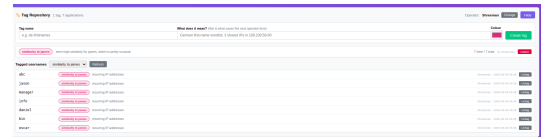


Figure 10: Show tag to other operators - Requirement R4

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